Atlas-Based Interior Tomography

Mona Selim, Essam A. Rashed and Hiroyuki Kudo

***Abstract*—Interior tomography is image reconstruction of a region-of-interest (ROI) located inside a scanned object. It has major effect in reducing the radiation dose. It was believed previously that the solution of interior problem was not unique. In the last decade, it has been learned that the interior problem can be solvable if sub-region inside the ROI is recognized or if the ROI is piecewise constant. Eventhough, such cases are not always available and the existence of recognized/currently known sub-region inside the ROI is not confirmed in many situations. In this paper, we introduce the use of prior knowledge from probabilistic atlas in solving the interior problem. The atlas is constructed by using a set of computed tomography (CT) images which are scanned in advance for different patients and fitted by Gaussian mixture model (GMM). Then, expectation maximization (EM) algorithm is used to estimate the mixture parameters. Whereas the probabilistic atlas is generated from advance scan of various patients, the prior information is always available and confirmed for all applications. We develop a statistical reconstruction method where the cost function is designed to incorporate the information of the probabilistic atlas. Both experimental results of the simulation and real data indicate that the proposed method remarkably improve the image quality and minimize the effect of DC-shift. Surprisingly, we observe a high quality reconstruction concerning the region outside the ROI.**

I. Introduction

Image reconstruction of region-of-interest (ROI) located inside a scanned object is called interior tomography or local tomography. In interior tomography, the projection data is limited because x-ray beams are directed only to ROI inside the object and the beam portion located left and right of the ROI is truncated in all view angles. Figure. 1 illustrates the configuration of ordinary CT and interior tomography. Interior tomography has great benefits in reducing the x-ray radiation dose to which the patient is exposed because it focuses the x-ray beams to a limited region of the whole object. Therefore, it is expected to decrease the risk to which the patient can be exposed, that is, regular CT scans may lead to a malignancy [1]. Also, it has many other benefits such as minimizing the necessary detector size, suppressing scatter artifact, allowing higher image magnification, handling large object and reducing system cost. Conventional analytical reconstruction, such as filtered back-projection (FBP) algorithm, for interior problem is known to result in DC-shift and low-frequency artifacts in the reconstructed image. In the last decade, several studies have investigated the interior problem in computed tomography. From these studies, it has been proven that the ROI can be accurately and stably reconstructed from the local projections if precise information for only small sub-region inside the ROI is recognized [2–4]. Also, based on compressed sensing theory, it has been confirmed that the solution of interior problem is achievable if the ROI is piecewise constant [5], [6]. Nevertheless, in many situations it is hard to get exact prior information of a sub-region and the ROI is not always piecewise constant.

This work was supported financially by the Science and Technology Development Fund (STDF), Egypt, Grant No 6104. The work of H. Kudo is supported by JSPS KAKENHI, Grant-in-Aid for Scientific Research (C), Grant Number JP15K06103. M. Selim is with the Dept. of Computer Science, Faculty of Computers and Information, Suez University, Suez, Egypt (E-mail: mona.selim@suezuniv.edu.eg). H. Kudo is with the Division of Information Engineering, Faculty of Engineering, Information and Systems, University of Tsukuba, Tsukuba, Japan and with the JST-ERATO, Momose Quantum-Beam Phase Imaging Project, Japan. E. A. Rashed is with the Dept. of Mathematics, Faculty of Science, Suez Canal University, Ismailia 41522, Egypt.

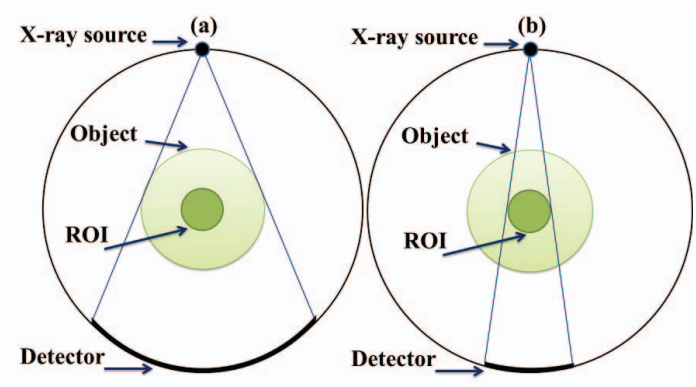

Fig. 1. Configuration of (a) ordinary CT and, (b) interior tomography.

In this paper, we introduce the use of prior knowledge from probabilistic atlas in solving the interior problem. This work is an extension of our previous work in [7]. The residual sections of this paper are organized as follow. Section II demonstrates briefly the approach used in the construction of the probabilistic atlas. Subsequently, the statistical image reconstruction method is proposed. Section III analyzes and displays the experimental results of simulation and real data. Section IV shows the conclusion of the paper.

II. Proposed Method

A. Pre-processing

The atlas is generated from a set of CT images of different patients scanned in advance. Two techniques of image processing are used to create the atlas: image registration and image segmentation. Image registration is the procedure of superimposing two or more images to have maximum correspondence between these images. The registration process can be analyzed into three components. First, the similarity measure for degree in two images match. Second, the transformation model, which define the approach to modifying the reference image to match the target image. A number of numerical parameters identify a particular instance of the transformation. Third, the optimization method that estimates the parameters of the transformation model to maximize the

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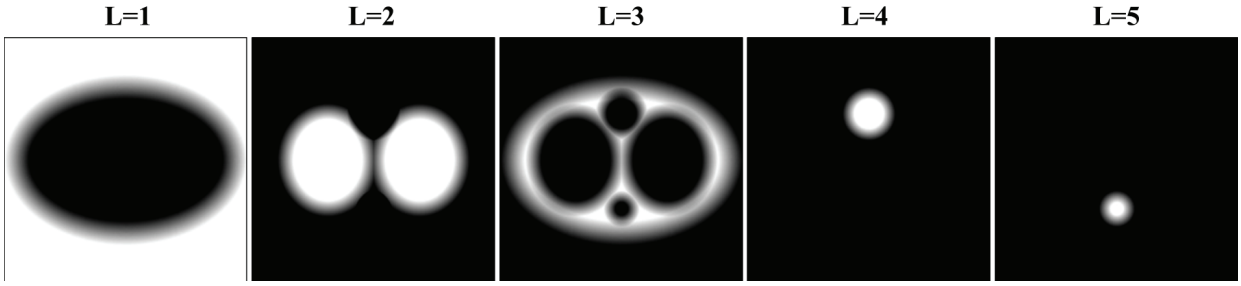

Fig. 2. Probabilistic atlas ($L = 5$) created from simulated 100 digital phantoms (white is 1 and black is 0 values).

matching criterion. After registration of the images in the dataset, the segmentation technique is used. Image segmentation means dividing the image pixels into categories or classes, each class corresponds to different parts of the object (image). The images are fitted using GMM [8] because it is characterized by simplicity, easy to implement, non-complex density function and requires small number of parameters. Then, EM algorithm [9] is used to estimate GMM mixture parameters.

We develop a statistical reconstruction method with the cost function is designed to incorporate the information of the probabilistic atlas. The major contribution of this paper is to ensure that using prior knowledge extracted from the atlas in the interior tomography produce higher image quality and minimize the effect of DC-shift. Surprisingly, we observe a high quality reconstruction concerning the region outside the ROI. The merit of using probabilistic atlas as prior information is to overcome the limitation of availability of known sub-region inside ROI. The atlas is constructed from old scan of numerous patients and this prior information is in most cases available.

B. Construction of the probabilistic atlas

The process of probabilistic atlas construction is performed through two main steps. The first step is the registration of the images in the dataset. We selected one individual image from the dataset and registered the remaining images on it by affine registration approach [10]. The transformation model used is affine which includes translation, rotation, scale and sheer parameters. The similarity metric and optimizer used for the registration process are mean square error and regular step gradient descent respectively. The second step is the segmentation of the registered images obtained from the first step using GMM. The dataset images are segmented into L clusters where each cluster is a composed represented by image pixels. The cluster denotes a single region (organ) of the patient. The probabilistic density function of GMM is defined as

$$p(x_j) = \sum_{l=1}^{L} \pi_l \; p(x_j|\Omega_l), \quad j = 1, \ldots, n \tag{1}$$

$$p(x_j|\Omega_l) = \frac{1}{\sqrt{2\pi_l \sigma_l^2}} \exp\left(-\frac{(x_j - \mu_l)^2}{2\sigma_l^2}\right), \tag{2}$$

where $\vec{x} = (x_1, \ldots, x_n)$ is the image vector. Each $p(x_j|\Omega_l)$ represents a component of the mixture with mean value μ_l and covariance σ_l. Ω_l, $(l = 1, \ldots, L)$ is the class labels. Also, π_l is the prior probability for every pixel x_j to belong to the class Ω_l and satisfy these conditions.

$$0 \leq \pi_l \leq 1 \quad \text{and} \quad \sum_{l=1}^{L} \pi_l = 1. \tag{3}$$

EM method is used to estimate the GMM parameter $(\pi_l, \mu_l, \lambda_l)$ and thus updates the posterior probability which is defined as

$$p(\Omega_l|x_j) = \frac{\pi_l p(x_j|\Omega_l)}{\sum_{l=1}^{L} \pi_l p(x_j|\Omega_l)}. \tag{4}$$

The parameters of GMM and the posterior probability are exploited as part of the cost function. Given the posterior probability, a reference image $\vec{u} = (u_1, \ldots, u_j)$ is computed as the class mean value with higher probability component as follow

$$u_j = \{\mu_{\hat{l}} \mid p(\Omega_{\hat{l}}|x_j) = \max_{l=1}^{L} \; p(\Omega_l|x_j)\} \tag{5}$$

C. Image reconstruction framework

The x-ray CT measurements can be depicted by the following statistical framework

$$y_i \approx \text{Poisson } [b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle)] \tag{6}$$

where $\vec{y} = (y_1, \ldots, y_m)$ represents the detector measurements with the blank scan $\vec{b} = (b_1, \ldots, b_m)$. $A = \{a_{ij}\}$ represents $m \times n$ system matrix. Image reconstruction algorithm depends on minimizing the following cost function

$$f(\vec{x}) = L(\vec{x}) + \vec{\gamma} D(\vec{x}, \vec{u}) \tag{7}$$

where $L(\vec{x})$ defines the negative log-likelihood function as follow

$$L(\vec{x}) = \sum_{i=1}^{m} [b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle) + y_i \langle \vec{a_i}, \vec{x} \rangle], \tag{8}$$

where $D(.,.)$ expresses the ℓ_1-norm distance and can be denoted by

$$D(\vec{x}, \vec{u}) = \sum_{j=1}^{n} d(x_j, u_j), \qquad d(a, b) = |a - b|. \tag{9}$$

$D(\vec{x}, \vec{u})$ is a distance function between the target image $\vec{x}$ and the reference image $\vec{u}$ computed from the atlas using Eq. (5).

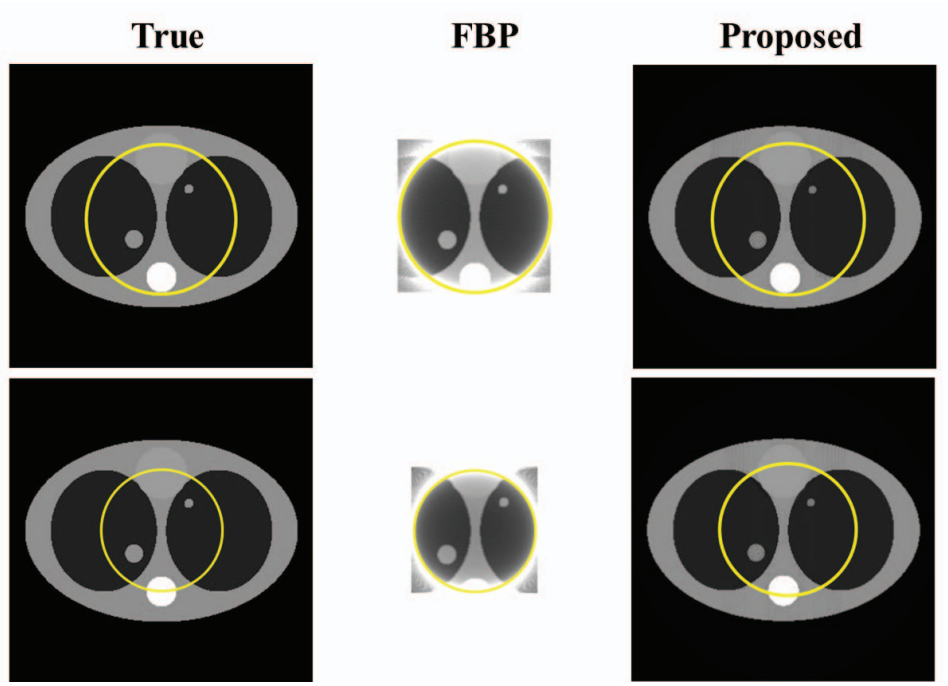

Fig. 3. The first column is the true phantom, the second and third are images reconstructed using FBP and the proposed method, respectively, with different sizes of ROI. Display gray scale is (0.0, 0.18) cm^{-1}.

The parameter $\vec{\gamma} = (\gamma_1, \ldots, \gamma_n)$ is probabilistic atlas weight where γ_j is defined as follow

$$\gamma_j = \max_{l=1}^{L} \; p(\Omega_l | x_j). \tag{10}$$

The parameter $\vec{\gamma}$ is computed as a pixel-dependent and is defined as the maximum posterior probability related with image pixel x_j as denoted in Eq. (10). The cost function in Eq. (7) is minimized using the majorization−minimization technique and the iterative reconstruction algorithm is defined similar to our previous work in [7]. The final iterative reconstruction algorithm can be mentioned as:

1 Initialization $x^{(0)} = \epsilon$, with $\epsilon > 0$, $k \leftarrow 0$ **repeat**
  for $j = 1 : n$ **do**
2 $p_j = x_j^k + x_j^k \frac{\sum_i a_{ij}\left[b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle) - y_i\right]}{\sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle)}$
3 $t_j = \gamma_j x_j^k / (\sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle))$
4 $x_j^{k+1} = Th(p_j, u_j, t_j)$
  end
  $k \leftarrow k + 1$
until *stopping criterion*;

where the thresholding function in step (4) is computed as

$$Th(a, b, c) = \begin{cases} a - c & (a > b + c) \\ a + c & (a < b - c) \\ b & (otherwise) \end{cases} . \tag{11}$$

III. Experimental results

A. Simulation data

We performed a simulation study to evaluate the proposed method. We used a set of 100 digital phantoms composed of five components to generate the atlas. Simulated organs are set to have small variations in size. The image size is 320 × 320 pixels with pixel size of 1 × 1 mm. The number of projections is 256 and the number of iterations is 50. To create the atlas, we picked out one image as reference and the other 99 images are registered onto the reference image. The EM algorithm is used to estimate the GMM parameters $(\pi_j, \mu_l, \lambda_l)$. Figure 2 shows the atlas constructed using simulated data. A digital phantom, with structures similar to the phantoms used in atlas generation and small lesion insert, is used to evaluate the proposed method. We have implemented FBP and the proposed method with different sizes of ROI. The results are shown in Fig. 3. Also, Fig. 4 shows the profile plot of horizontal line through the true and reconstructed images using FBP in addition to the proposed method. It is clear from these results that the proposed method achieve higher image quality compared to FBP reconstruction and minimizes DC-shift artifacts. Also, the lesion inside ROI is reconstructed accurately although it is not found in the atlas.

B. Real data

In order to evaluate the proposed method, we created the atlas using dataset of screening chest CT volumes for 68 patients. Every volume has 25 to 31 slices and the slice size is 320×320 pixels with pixel size of 1×1 mm. We selected 55 volumes to generate the atlas. One volume is chosen as a reference and the residual 54 volumes are registered onto the reference volume. Figure 5 demonstrates the created atlas which composed of 6 classes (bone, muscles, fatty-tissues, lungs, air and bed). We used central slice of one volume not involved in the process of atlas construction to evaluate the proposed method. The number of projection views is 256

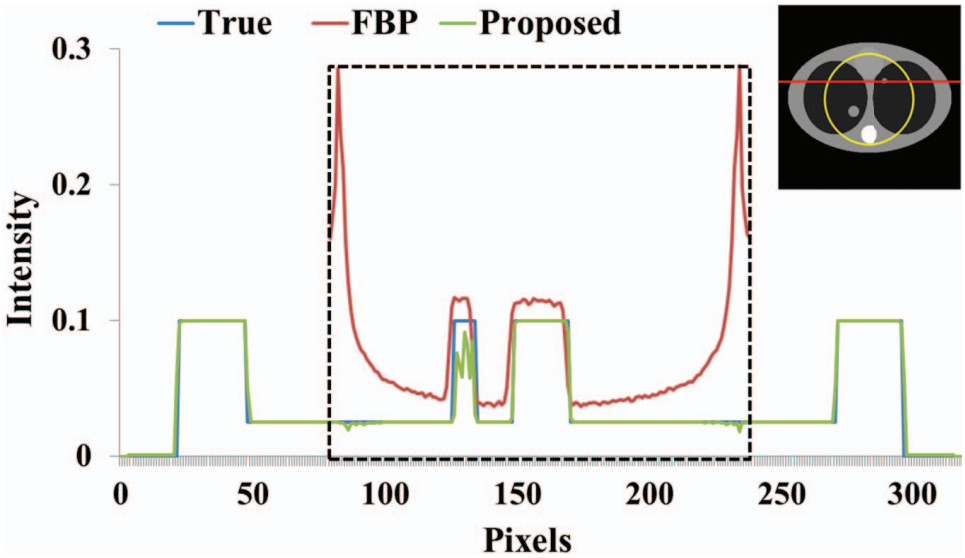

Fig. 4. Profile plot of red line (shown in the top right image) through the true, and reconstructed images using FBP and the proposed method.

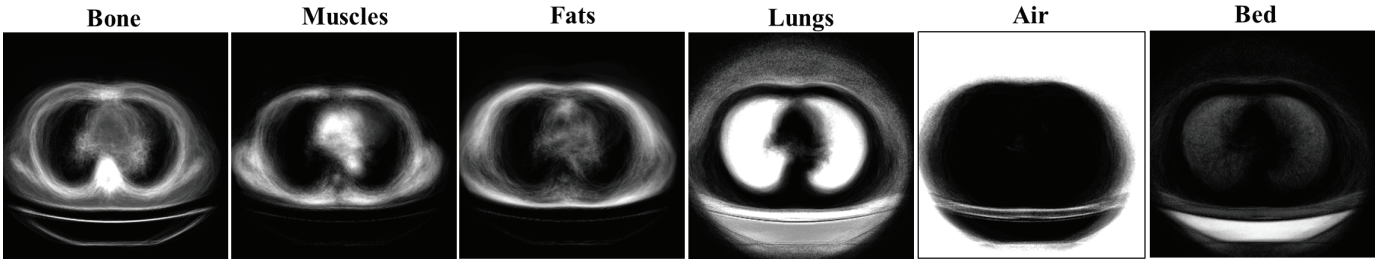

Fig. 5. Probabilistic atlas ($L = 6$) created from chest screening CT images of 55 patients using Gaussian mixture model (white is 1 and black is 0 values).

the number of iterations are 150 iterations. We executed FBP and the proposed algorithm with different sizes of ROI. The results are shown in Fig 6. Also, Fig. 7 shows profile plot of horizontal line through the true and reconstructed images using FBP in addition to the proposed method. From the results, it can be observed that using the probabilistic atlas in image reconstruction of interior tomography gives better image quality than FBP algorithms and reduces effectively DC-shift artifacts.

IV. Conclusion

This paper investigates the use of prior information obtained from probabilistic atlas in solving interior problem in computed tomography. The proposed method can effectively reduce the DC-shift and low-frequency artifacts. Using probabilistic atlas as prior information in solving the interior problem overcomes the limitation of availability of known subregion inside ROI in many cases where the atlas is obtained from old scan of many patients. The interesting point we observe from these results is that this prior can preserve abnormalities, such as lesions, and effectively retrieve a large area outside the ROI. We have never observe a similar results in the past attempts to solve the interior problem.

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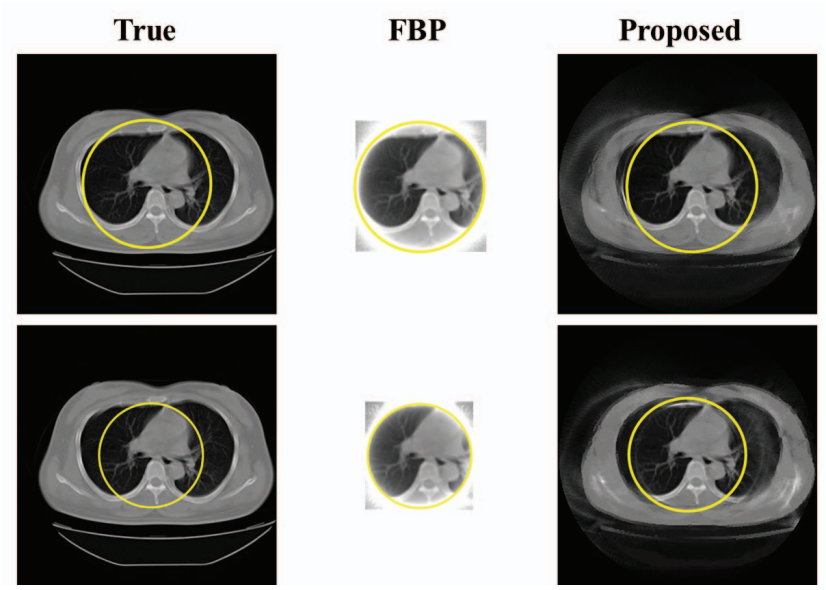

Fig. 6. Central slice of patient number 12 of the chest CT dataset. The first column is the original slice image, the second, third and third are image reconstructed using FBP and the proposed method with different sizes of ROI. Display gray scale is (0.08, 1.93) cm^{-1}.

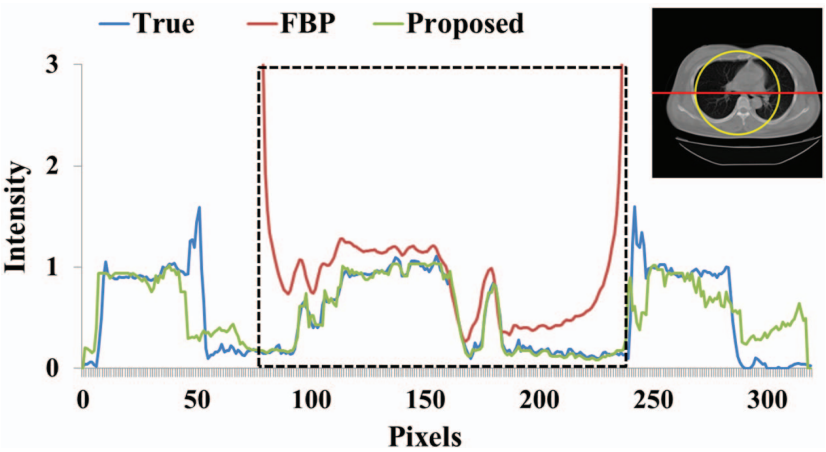

Fig. 7. Profile plot of red line (shown in the top right image) through the true and reconstructed images using FBP and the proposed method.